

On Homology of Jordan Algebras

1 Introduction

For Lie algebras, homology is classically defined by the Chevalley–Eilenberg complex and admits several equivalent interpretations: as the derived functor of coinvariants, as a Tor-group over the universal enveloping algebra, and, in the homotopical setting, as derived abelianization. For Jordan algebras, the situation is less uniform. There is no single homology theory that has become as canonical as Lie algebra homology. Instead, several related approaches exist.

The purpose of this note is to explain a natural modern definition of homology for Jordan algebras using operads and derived abelianization, and to compare it with the older Lie-theoretic approach via the Tits–Kantor–Koecher construction.

2 Jordan algebras and the problem of homology

Let k be a field of characteristic 0. A Jordan algebra over k is a commutative (generally nonassociative) algebra J satisfying the Jordan identity

$$(x^2y)x = x^2(yx).$$

Since Jordan algebras are already commutative, the usual associative or Lie-theoretic notion of abelianization cannot be imported directly by quotienting by commutators.

For this reason, the correct analogue of “abelian objects” in the category of Jordan algebras is the class of Jordan algebras with *zero multiplication*. Thus the relevant “abelianization” functor is not

$$J \mapsto J/[J, J],$$

but rather the functor of *indecomposables*

$$Q(J) := J/J^2,$$

where J^2 denotes the linear span of all products xy with $x, y \in J$.

3 Operadic viewpoint

Over a field of characteristic 0, Jordan algebras are algebras over the Jordan operad, which we denote by Jord . More precisely, Loday–Vallette explain that although the usual Jordan identity is not multilinear, when 3 is invertible it is equivalent to its multilinearized cubic form; they then define the associated operad and denote it by Jord [?, §13.10.1, p. 437]. In the general operadic framework, multilinear identities in characteristic 0 determine an operad and the resulting category of algebras is equivalent to the original category of algebras of that type [?, Proposition 5.6.4].

Therefore dg Jordan algebras may be regarded as algebras over Jord in the symmetric monoidal category Ch_k of chain complexes over k .

4 Model category structure on dg Jordan algebras

Write

$$\mathrm{dgJord}_k := \mathrm{Alg}_{\mathrm{Jord}}(\mathrm{Ch}_k).$$

To put a homotopy theory on this category, one uses the transferred model structure for algebras over operads.

A general admissibility theorem of Pavlov–Scholbach states that if $\mathcal{C}$ is a suitable symmetric monoidal model category, then for any symmetric colored operad O , the category $\mathrm{Alg}_O(\mathcal{C})$ admits a transferred model structure whose weak equivalences and fibrations are created in $\mathcal{C}$ [?, Theorem 1.1]. In the case of chain complexes over a field of characteristic 0, one can cite a more concrete statement: White–Yau prove that if R is a semisimple commutative ring containing $\mathbb{Q}$, then every colored operad in chain complexes over R is admissible; more precisely, for such R , every operad is Σ -cofibrant and their Theorem 4.1.1 implies that all operads are admissible [?, Corollary 5.2.4]. Since a field of characteristic 0 is semisimple and contains $\mathbb{Q}$, this applies to k .

Hence dgJord_k inherits a transferred model structure from Ch_k . In particular, weak equivalences and fibrations are detected on the underlying complexes.

5 Derived abelianization

Let

$$Q : \mathrm{dgJord}_k \longrightarrow \mathrm{Ch}_k, \quad Q(J) = J/J^2,$$

be the functor of indecomposables. This functor is left adjoint to the inclusion

$$\mathrm{Ch}_k \hookrightarrow \mathrm{dgJord}_k$$

that sends a complex V to the same complex equipped with zero Jordan product.

Now let J be a dg Jordan algebra, and choose a cofibrant replacement

$$\tilde{J} \xrightarrow{\sim} J$$

in dgJord_k . The *derived abelianization* of J is then defined by

$$\mathbb{L}Q(J) := Q(\tilde{J}) = \tilde{J}/(\tilde{J})^2.$$

Its homology

$$H_n^{\mathrm{Jord}}(J) := H_n(\mathbb{L}Q(J))$$

may be regarded as the *Quillen homology* of the Jordan algebra J .

This gives a natural homology theory for Jordan algebras, entirely parallel to Quillen homology for associative, commutative, and Lie algebras.

6 Relation with the Tits–Kantor–Koecher construction

Another natural route is to attach to a Jordan algebra, Jordan pair, or Jordan triple a Lie algebra by means of the Tits–Kantor–Koecher (TKK) construction. One may then study Lie algebra homology of the resulting TKK Lie algebra.

This produces a useful and often computable invariant. However, it is conceptually different from the intrinsic operadic homology of Jordan algebras defined above. The TKK approach translates Jordan theory into Lie theory, whereas derived abelianization remains entirely internal to the category of Jordan algebras.

7 Conclusion

There is no universally established classical homology theory of Jordan algebras completely analogous to Chevalley–Eilenberg homology for Lie algebras. Nevertheless, the operadic viewpoint provides a clean and natural definition. Namely, one regards dg Jordan algebras as algebras over the Jordan operad, equips their category with the transferred model structure, and defines homology as the left derived functor of indecomposables:

$$H_n^{\text{Jord}}(J) = H_n(\mathbb{L}(J \mapsto J/J^2)).$$

This may be called the homology of Jordan algebras in the sense of Quillen.

References

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